

PROJECT REPORT

Custom Network Proxy Server

A production-grade forward proxy server built with Python's socket programming — featuring HTTP/HTTPS tunneling, PBKDF2 authentication, LRU caching, rate limiting, and real-time monitoring.

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Domain: Technology — Software Development & Engineering Research

Language: Python 3.10+ (Zero External Dependencies)

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🔒 PBKDF2 Auth

📦 LRU Cache

🛡️ Rate Limiting

📊 Live Dashboard

🧪 43 Tests

📄 RFC 7230

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1. Abstract

This report presents the design, implementation, and evaluation of a **production-grade forward proxy server** built entirely using Python's standard library. The proxy server intercepts, inspects, and forwards HTTP and HTTPS traffic between clients and origin servers, implementing enterprise-level features including per-IP rate limiting using the Token Bucket algorithm, PBKDF2-HMAC-SHA256 cryptographic authentication, thread-safe LRU response caching with TTL eviction, domain/IP/CIDR blocklist filtering with hot-reload, structured JSON logging with rotation, and a real-time monitoring dashboard with Server-Sent Events (SSE). The entire system operates with **zero external dependencies** and is validated by a comprehensive test suite of **43 automated tests** covering unit, integration, and concurrency scenarios.

13SOURCE
MODULES**43**AUTOMATED
TESTS**0**EXTERNAL
DEPS**6**

PIPELINE STEPS

RFC7230
COMPLIANT

2. Introduction

2.1 Problem Statement

In modern networked environments, proxy servers serve as critical intermediaries between clients and the internet, providing security enforcement, access control, performance optimization, and traffic monitoring. However, most existing proxy solutions are either:

- **Opaque commercial products** (Squid, HAProxy, Nginx) — powerful but not designed for understanding underlying network programming concepts
- **Simplistic educational examples** — basic socket demos that handle only GET requests without any production considerations

There exists a significant gap between textbook TCP/IP socket programming examples and real-world proxy server implementations. This project bridges that gap by building a **fully-functional, production-grade proxy server from scratch** that demonstrates mastery of networking, security, concurrency, and systems programming.

2.2 Motivation

The project is motivated by three key drivers:



Deep Learning

Understanding TCP/IP, HTTP protocol internals, and RFC specifications by implementing them from the ground up rather than using high-level abstractions.



Security Engineering

Implementing cryptographic authentication, rate limiting, and access control — critical skills for building secure networked systems.



Concurrent Systems

Solving real concurrency challenges: thread pools, lock granularity, shared-state management, and graceful shutdown under load.



Observability

Building production-grade monitoring: structured logging, real-time dashboards, and metrics collection — essential for operating networked services.

2.3 Objectives

The primary objectives of this project are:

1. **HTTP/HTTPS Proxy:** Implement full HTTP request forwarding and HTTPS CONNECT tunneling per RFC 7230
2. **Security:** Implement PBKDF2-HMAC-SHA256 authentication with constant-time comparison, per-IP rate limiting, and domain/IP filtering
3. **Performance:** Implement bounded thread pool concurrency and LRU response caching with TTL
4. **Observability:** Build structured JSON logging with rotation and a real-time web monitoring dashboard

5. **Quality:** Achieve comprehensive test coverage with 43+ automated tests across all modules
6. **Zero Dependencies:** Build the entire system using only Python's standard library

3. Technology Stack

 Key Design Decision: Zero External Dependencies

The entire proxy server is built using only Python's standard library. This deliberate constraint demonstrates deep knowledge of Python's built-in capabilities and ensures the project can run on any Python 3.10+ installation without `pip install`.

Layer	Technology	Standard Library Module	Purpose
Networking	TCP Sockets	<code>socket</code>	Low-level TCP connection management
Concurrency	Thread Pool	<code>concurrent.futures</code>	Bounded worker threads for parallel request handling
Synchronization	Locks	<code>threading</code>	Thread-safe access to shared resources
Cryptography	PBKDF2-HMAC-SHA256	<code>hashlib</code> , <code>hmac</code> , <code>os</code>	Password hashing with 600K iterations
Caching	OrderedDict LRU	<code>collections</code>	O(1) LRU cache with TTL eviction
IP Handling	CIDR Networks	<code>ipaddress</code>	IPv4/IPv6 address and network matching
Logging	Rotating File Logger	<code>logging</code> , <code>logging.handlers</code>	Structured JSON logging with size-based rotation
Configuration	INI Parser	<code>configparser</code>	Typed configuration with defaults

Layer	Technology	Standard Library Module	Purpose
Dashboard	HTTP Server	<code>http.server</code> , <code>socketserver</code>	Real-time monitoring web interface
URL Parsing	URL Parser	<code>urllib.parse</code>	HTTP request URI decomposition
Testing	Unit Tests	<code>unittest</code>	43 automated test cases
Dashboard Frontend	Chart.js (CDN)	—	Real-time line charts (loaded via CDN in browser)

Language: Python 3.10+ (required for `type | None` union syntax and `match` statement support)

Platform: Linux, macOS, WSL (Windows Subsystem for Linux)

4. System Architecture

4.1 High-Level Architecture

The proxy follows a **modular, layered architecture** where each component has a single responsibility. The system is designed around a central **6-step request pipeline** that processes every incoming connection through parsing, rate limiting, authentication, filtering, and forwarding stages.

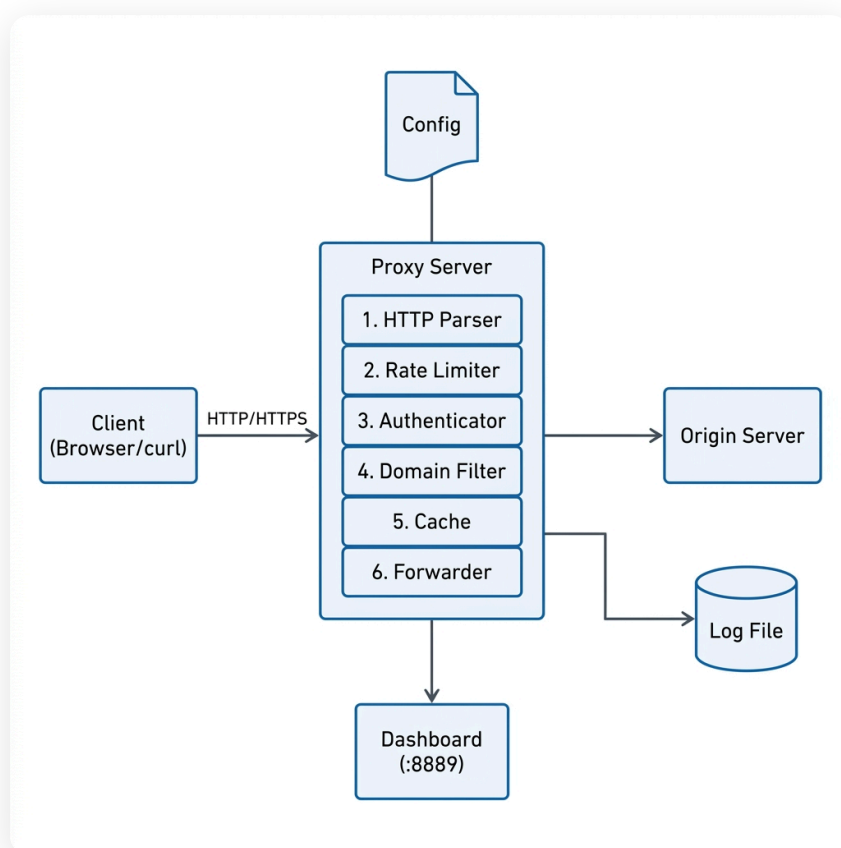


Figure 1: High-level system architecture showing the modular component layout with request flow from client through the proxy pipeline to origin servers, and the parallel monitoring dashboard.

4.2 Project Structure

Custom-Network-Proxy-Server/

- └─ **src/** ← 13 source modules (~90 KB)
 - └─ **server.py** ← Entry point: TCP socket + ThreadPoolExecutor
 - └─ **handler.py** ← 6-step request pipeline orchestrator
 - └─ **parser.py** ← HTTP request parser (RFC 7230 §3)
 - └─ **forwarder.py** ← HTTP forwarding + HTTPS CONNECT tunneling
 - └─ **filter.py** ← Domain/IP/CIDR blocklist with hot-reload
 - └─ **cache.py** ← Thread-safe LRU cache with TTL eviction
 - └─ **logger.py** ← Structured JSON logging + rotation + ring buffer
 - └─ **log_schema.py** ← Canonical LogEntry dataclass for all events
 - └─ **rate_limiter.py** ← Per-IP token bucket rate limiting
 - └─ **auth.py** ← PBKDF2-HMAC-SHA256 password hashing
 - └─ **config_loader.py** ← INI config parser with typed defaults
 - └─ **dashboard.py** ← Real-time monitoring HTTP server (SSE)
 - └─ **dashboard_ui.py** ← Self-contained dashboard HTML/CSS/JS
- └─ **config/** ← Configuration files
 - └─ **proxy.conf** ← Server settings (INI format)
 - └─ **blocked_domains.txt** ← Domain/IP blocklist
 - └─ **users.txt** ← PBKDF2-hashed credentials
- └─ **tests/** ← 43 automated tests + shell smoke tests
- └─ **tools/** ← CLI user management utility
- └─ **docs/** ← Architecture documentation
- └─ **logs/** ← Rotating log output directory

4.3 Component Overview

Module	Lines	Responsibility
<code>server.py</code>	105	TCP socket listener, thread pool creation, signal handler registration, graceful shutdown
<code>handler.py</code>	240	6-step request pipeline orchestration — parse, rate limit, auth, filter, tunnel/forward
<code>parser.py</code>	194	HTTP request parsing: header accumulation loop, absolute/origin URI, Content-Length body
<code>forwarder.py</code>	318	HTTP forwarding with cache-aside pattern, HTTPS CONNECT bidirectional relay

Module	Lines	Responsibility
<code>filter.py</code>	123	Domain/IP/CIDR blocklist parsing, matching, and hot-reload
<code>cache.py</code>	184	OrderedDict LRU cache with TTL, CacheStats for observability
<code>auth.py</code>	116	PBKDF2-HMAC-SHA256 hashing (600K iterations), constant-time verification
<code>rate_limiter.py</code>	208	Per-IP token bucket with stale bucket pruning, fine-grained locking
<code>logger.py</code>	215	Dual-output logging (JSON file + human stderr), ring buffer for dashboard
<code>log_schema.py</code>	119	Canonical LogEntry dataclass — single source of truth for log format
<code>config_loader.py</code>	89	INI config parser with typed defaults for all 5 subsystems
<code>dashboard.py</code>	194	HTTP server with SSE streaming, metrics/logs API endpoints
<code>dashboard_ui.py</code>	~600	Self-contained HTML/CSS/JS with glassmorphism UI and Chart.js

5. Design & Implementation

5.1 Server Core & Concurrency Model

The server uses a **bounded ThreadPoolExecutor** pattern instead of the naive thread-per-connection approach. This prevents resource exhaustion under load while maintaining high concurrency.

Why ThreadPoolExecutor over thread-per-connection?

An unbounded thread-per-connection model creates a new OS thread for every client. Under load (or attack), this can exhaust system resources (file descriptors, memory). A bounded pool with 20 workers caps resource usage at a known limit. Excess connections queue in the executor, and the OS `listen(50)` backlog provides a secondary bound.

Server Initialization Code

```
def start_proxy():  
    """Bind, listen, and accept connections in the main thread."""  
    # Create listening TCP socket  
    _server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)  
    _server_socket.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)  
    _server_socket.bind((LISTEN_HOST, LISTEN_PORT))  
    _server_socket.listen(MAX_CONN)  
  
    # Create bounded thread pool (20 workers)  
    _thread_pool = ThreadPoolExecutor(  
        max_workers=POOL_SIZE,  
        thread_name_prefix="proxy-worker",  
    )  
  
    # Start dashboard (daemon thread)  
    start_dashboard()  
  
    # Accept loop — main thread blocks here  
    while True:  
        client_sock, client_addr = _server_socket.accept()  
        _thread_pool.submit(handle_client, client_sock, client_addr)
```

Graceful Shutdown

Signal handlers (`SIGINT` , `SIGTERM`) trigger a clean shutdown sequence:

1. Close the listening socket → accept loop exits
2. `ThreadPoolExecutor.shutdown(wait=True)` → drains in-flight requests
3. Dashboard daemon thread auto-terminates (marked as daemon)
4. Zero resource leaks — all sockets closed, all threads joined

5.2 Request Pipeline

Every client connection passes through a strict **6-step pipeline** in `handler.py`. Each step either advances the request to the next stage or terminates it with an appropriate HTTP error response. This design ensures consistent enforcement of all security policies.

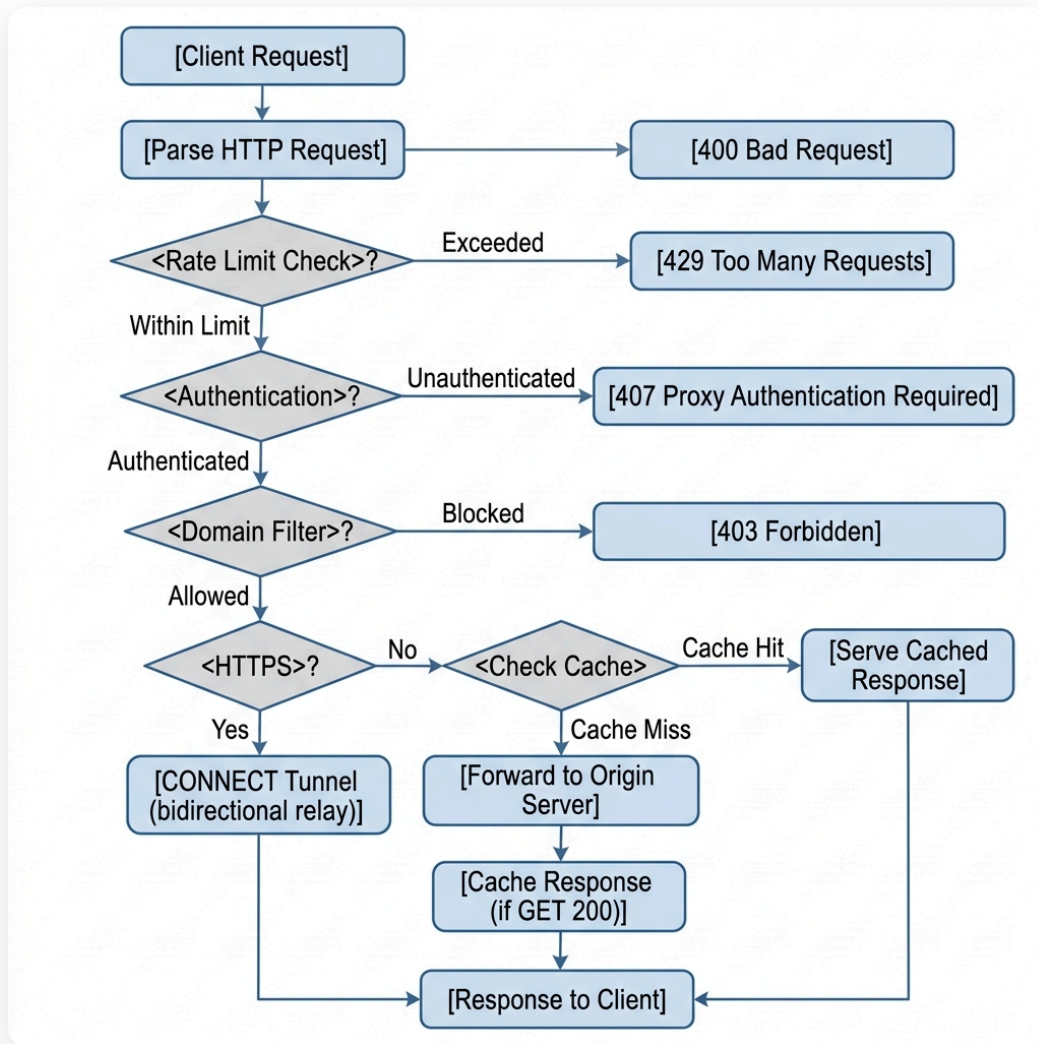


Figure 2: Complete request processing pipeline showing all 6 steps from client request to response, with failure paths and HTTP error codes at each stage.

Pipeline Steps in Detail

1

Parse HTTP Request

Receive raw bytes from the client socket, accumulate until `\r\n\r\n` header terminator is found, then parse method, URI, headers, and body. Supports absolute-URI, origin-form, and CONNECT request types.

🚫 400 Bad Request on malformed input

2

Rate Limit Check (Token Bucket)

Check the client's IP against its token bucket. If the bucket is empty (no tokens available), reject immediately with `Retry-After` header. **This runs before authentication** to provide fail-fast protection against brute-force attacks.

🚫 429 Too Many Requests + Retry-After header

3

Authentication

Decode the `Proxy-Authorization: Basic <base64>` header, split into username:password, look up the stored PBKDF2 hash, and verify using constant-time comparison.

🚫 407 Proxy Authentication Required

4

Domain / IP Filtering

Check the destination host against the blocklist. Supports exact domain match, subdomain suffix match, exact IP, and CIDR range matching. Blocklist is hot-reloadable at runtime.

🚫 403 Forbidden

5

HTTPS CONNECT Tunneling

For `CONNECT` requests: establish a TCP connection to the destination, send `200 Connection Established`, then relay bytes bidirectionally using two daemon threads. The proxy **never inspects encrypted TLS content**.

🚫 502 Bad Gateway on connection failure

6

HTTP Forwarding (with Cache-Aside)

For HTTP requests: check the LRU cache first. On miss, connect to the origin server, rebuild the request (strip hop-by-hop headers, inject `Via` and `X-Forwarded-For`), forward, and cache GET 200 responses.

🚫 502 Bad Gateway on origin failure

5.3 HTTP Parsing (RFC 7230)

The parser in `parser.py` implements HTTP/1.1 request parsing per **RFC 7230 §3**. It handles the complexities of real-world HTTP traffic that simplistic parsers miss.

Header Accumulation Loop

TCP is a stream protocol — data arrives in arbitrary-sized chunks. The parser cannot assume a single `recv()` call will return the complete HTTP headers. Instead, it uses an accumulation loop:

```
def recv_http_request(sock):
    """Read until full HTTP header block is received."""
    data = b""
    sock.settimeout(30)

    # Phase 1: accumulate headers until \r\n\r\n
    while HEADER_TERMINATOR not in data:
        if len(data) > MAX_HEADER_SIZE: # 64 KB guard
            return b""
        chunk = sock.recv(BUFFER_SIZE)
        if not chunk: break
        data += chunk

    # Phase 2: read body if Content-Length present
    content_length = _extract_content_length(header_block)
    if content_length and content_length > 0:
        while remaining > 0:
            chunk = sock.recv(min(BUFFER_SIZE, remaining))
            data += chunk
            remaining -= len(chunk)

    return data
```

Three Request Forms Handled

Form	Example	When Used
Absolute URI	<code>GET http://example.com/path?q=1</code> <code>HTTP/1.1</code>	Standard proxy request format

Form	Example	When Used
Origin Form	GET /path HTTP/1.1 + Host: example.com	Transparent proxy or direct connection
CONNECT	CONNECT example.com:443 HTTP/1.1	HTTPS tunneling

Each request is annotated with a **UUID4 request ID** for end-to-end log correlation, enabling operators to trace a single request across all log entries.

5.4 HTTP Forwarding & HTTPS Tunneling

HTTP Forwarding Path

When forwarding HTTP requests, the proxy rebuilds the outbound request to comply with RFC 7230:

URI Rewriting

Converts absolute URI (`GET http://host/path`) to origin-form (`GET /path`) per RFC 7230 §5.3.1

Header Stripping

Removes 9 hop-by-hop headers:
`Connection` , `Keep-Alive` , `Proxy-Authorization` , `TE` , `Trailers` , `Transfer-Encoding` , `Upgrade` , `Proxy-Authenticate` , `Proxy-Connection`

Header Injection

Injects `X-Forwarded-For` (client IP chain) and `Via: 1.1 proxy` per RFC 7230 §5.7.1

Body Forwarding

Forwards request body for POST, PUT, and PATCH methods using `Content-Length`

Outbound Request Construction

```
def _build_outbound_request(parsed: dict) → bytes:
    # Convert to origin-form
    request_line = f"{method} {path} HTTP/1.1\r\n"

    # Filter headers — skip hop-by-hop
    for key, value in headers.items():
        if key in HOP_BY_HOP_HEADERS: continue
        if key == "host": continue
        out_headers.append(f"{key}: {value}")

    # Inject proxy headers
    out_headers.append("Connection: close")
    out_headers.append(f"X-Forwarded-For: {client_ip}")
    out_headers.append("Via: 1.1 proxy")

    # Append body for POST/PUT/PATCH
    if method in ("POST", "PUT", "PATCH") and body:
        request_bytes += body

    return request_bytes
```

HTTPS CONNECT Tunneling

For HTTPS requests, the proxy establishes a **bidirectional TCP tunnel** between the client and the origin server. Two daemon threads relay bytes in opposite directions. The proxy **intentionally never inspects TLS content** — this is a deliberate security decision.

```
def tunnel(client_sock, server_sock):  
    """Bidirectional byte relay using two daemon threads."""  
    def _forward(src, dst):  
        while True:  
            data = src.recv(BUFFER_SIZE)  
            if not data: break  
            dst.sendall(data)  
  
    t1 = Thread(target=_forward, args=(client_sock, server_sock))  
    t2 = Thread(target=_forward, args=(server_sock, client_sock))  
    t1.start(); t2.start()  
    t1.join(timeout=120); t2.join(timeout=120)
```

5.5 Authentication — PBKDF2-HMAC-SHA256

The authentication system implements **PBKDF2-HMAC-SHA256**, a password hashing algorithm recommended by **NIST SP 800-132** and **OWASP 2024**. This is the same class of algorithm used by major systems like Django, Atlassian, and Cisco.

Security Property	Implementation	Rationale
Algorithm	PBKDF2-HMAC-SHA256	NIST SP 800-132 recommendation
Iterations	600,000	OWASP 2024 minimum for SHA-256
Salt	32-byte random (per user)	Prevents rainbow table attacks
Key Length	32 bytes	Matches SHA-256 output size
Comparison	<code>hmac.compare_digest()</code>	Constant-time — prevents timing attacks
Storage Format	<code>pbkdf2\$600000\$salt_hex\$hash_hex</code>	Self-describing, includes iteration count

Password Hashing Implementation

```
def hash_password(plain: str) → str:
    """Hash a plaintext password using PBKDF2-HMAC-SHA256."""
    salt = os.urandom(32) # 32-byte random salt
    dk = hashlib.pbkdf2_hmac(
        "sha256",
        plain.encode("utf-8"),
        salt,
        600_000, # OWASP 2024 recommended iterations
        dklen=32,
    )
    return f"pbkdf2${600000}${salt.hex()}${dk.hex()}"

def verify_password(plain: str, stored: str) → bool:
    """Verify using constant-time comparison."""
    _, iterations_str, salt_hex, hash_hex = stored.split("$")
    dk = hashlib.pbkdf2_hmac(
        "sha256", plain.encode(), bytes.fromhex(salt_hex),
        int(iterations_str), dklen=len(bytes.fromhex(hash_hex))
    )
    return hmac.compare_digest(dk, bytes.fromhex(hash_hex))
```

User Management CLI

A companion CLI tool (`tools/manage_users.py`) provides user administration:

```
# Add a user (password is automatically PBKDF2-hashed)
$ python tools/manage_users.py add alice mypassword
[+] Added user: alice

# List users with credential type
$ python tools/manage_users.py list
Username          Credential Type
-----
admin             PBKDF2 (hashed)
user              PBKDF2 (hashed)

# Migrate plaintext passwords to PBKDF2
$ python tools/manage_users.py migrate
[+] Migrated: legacy_user
[+] Migrated 1 user(s) to PBKDF2.
```

5.6 Domain & IP Filtering

The filtering module provides configurable access control based on domains, IP addresses, and CIDR ranges. The blocklist is loaded once at startup into memory for O(1) lookups and supports **hot-reload** at runtime without restarting the proxy.

Supported Filter Types

Type	Example	Matches
Exact Domain	example.com	example.com
Subdomain Suffix	example.com	sub.example.com , a.b.example.com
Exact IP	192.0.2.5	192.0.2.5
CIDR Range	192.168.0.0/24	All 256 IPs in 192.168.0.0-192.168.0.255

Blocklist Configuration

```
# config/blocked_domains.txt
# One entry per line. Blank lines and # comments ignored.

# Blocked domains (also blocks subdomains)
example.com
badsite.org

# Blocked IPs
192.0.2.5

# Blocked CIDR ranges
10.0.0.0/8
```

Matching Algorithm

```
def is_blocked(host: str) → bool:
    host = host.lower().strip()

    # 1. Exact domain match
    if host in _blocked_domains: return True

    # 2. Subdomain suffix match
    for blocked in _blocked_domains:
        if host.endswith("." + blocked): return True

    # 3. IP / CIDR matching
    addr = ipaddress.ip_address(host)
    for network in _blocked_networks:
        if addr in network: return True

    return False
```

5.7 Response Caching — LRU with TTL

The caching module implements an **in-memory Least-Recently-Used (LRU) cache** with **time-to-live (TTL) eviction**. It follows the **cache-aside pattern**: the application checks the cache before contacting the origin server, and stores responses after receiving them.

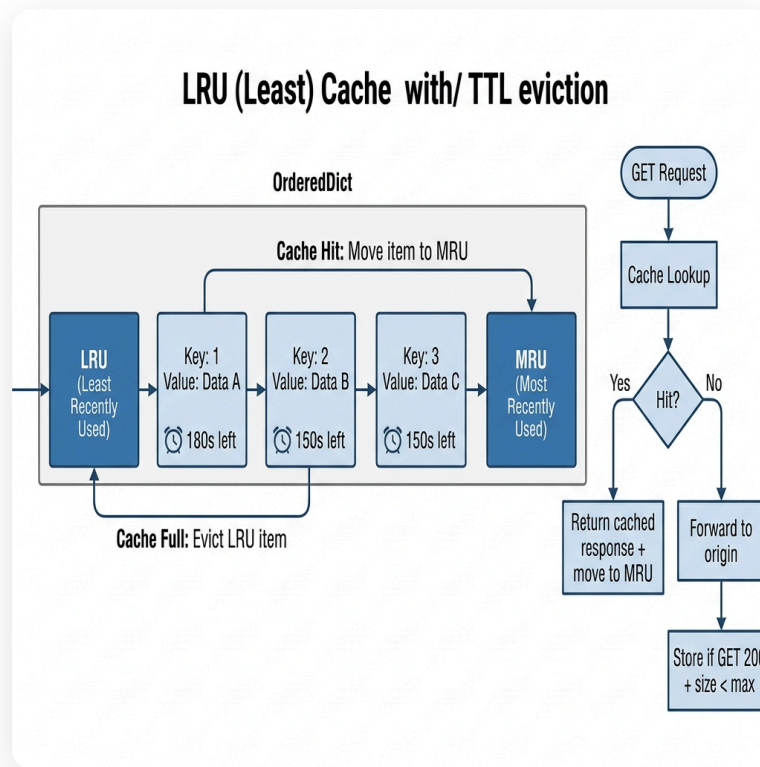


Figure 3: LRU Cache architecture showing *OrderedDict*-based eviction, TTL expiration, and the cache-aside access pattern for HTTP GET requests.

Cache Design Decisions

Decision	Choice	Rationale
Data Structure	<code>collections.OrderedDict</code>	$O(1)$ insert, lookup, delete, and LRU reordering
TTL Strategy	Lazy expiration on <code>get()</code>	No background sweeper thread — reduces complexity
Cacheable Requests	GET only	POST/PUT/PATCH have side effects

Decision	Choice	Rationale
Cacheable Responses	200 OK only	Error/redirect responses shouldn't be cached
Size Limit	512 KB per object	Prevents large downloads from dominating cache
Thread Safety	<code>threading.Lock</code>	Single lock for all mutations

Cache Implementation

```
class LRUCache:
    def get(self, key):
        with self.lock:
            if key not in self.cache:
                self.stats.record_miss()
                return None

            entry = self.cache[key]

            # TTL expiry check
            age = time.time() - entry["timestamp"]
            if age > self.ttl:
                del self.cache[key]
                self.stats.record_eviction(entry["size"])
                return None

            # Move to MRU end
            self.cache.move_to_end(key)
            self.stats.record_hit()
            return entry

    def put(self, key, value):
        with self.lock:
            self.cache[key] = value
            # Evict LRU if over capacity
            while len(self.cache) > self.capacity:
                _, evicted = self.cache.popitem(last=False)
                self.stats.record_eviction(evicted["size"])
```

Cache Observability

The `CacheStats` class tracks operational metrics with atomic thread-safe counters, exposed via the dashboard API:



5.8 Rate Limiting — Token Bucket Algorithm

Rate limiting is implemented using the **Token Bucket algorithm**, a standard approach used by AWS API Gateway, Nginx, and Cloudflare. Each unique client IP address gets its own token bucket that refills at a configurable rate.

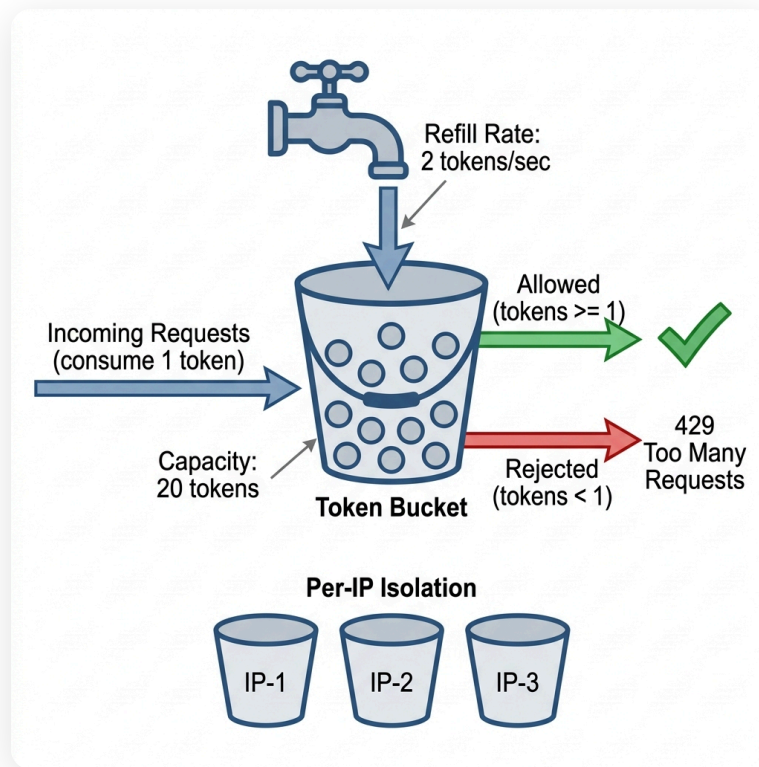


Figure 4: Token Bucket algorithm showing per-IP isolation, token consumption on each request, continuous refill, and rejection when the bucket is empty.

Algorithm Parameters

Parameter	Default	Description
Capacity	20 tokens	Maximum burst size — allows 20 rapid-fire requests
Refill Rate	2 tokens/sec	Sustained request rate when bucket isn't full
Stale Threshold	600 seconds	Prune buckets inactive for >10 minutes

Token Bucket Implementation

```
class TokenBucket:
    def consume(self) → bool:
        """Try to consume one token. Returns True if allowed."""
        with self.lock:
            now = time.monotonic()
            elapsed = now - self.last_refill

            # Refill tokens based on elapsed time
            self.tokens = min(
                self.capacity,
                self.tokens + elapsed * self.refill_rate,
            )
            self.last_refill = now

            if self.tokens ≥ 1.0:
                self.tokens -= 1.0
                return True
            return False
```

⚠ Design Decision: Rate Limiting Before Authentication

Rate limiting is checked **before** authentication (Step 2 vs Step 3 in the pipeline). This is a deliberate security design: it provides **fail-fast protection against brute-force attacks**. Without this ordering, an attacker could send millions of authentication attempts per second, overloading the expensive PBKDF2 computation (600K iterations per attempt).

Fine-Grained Locking Strategy

The rate limiter uses a **two-tier locking strategy** to minimize contention:

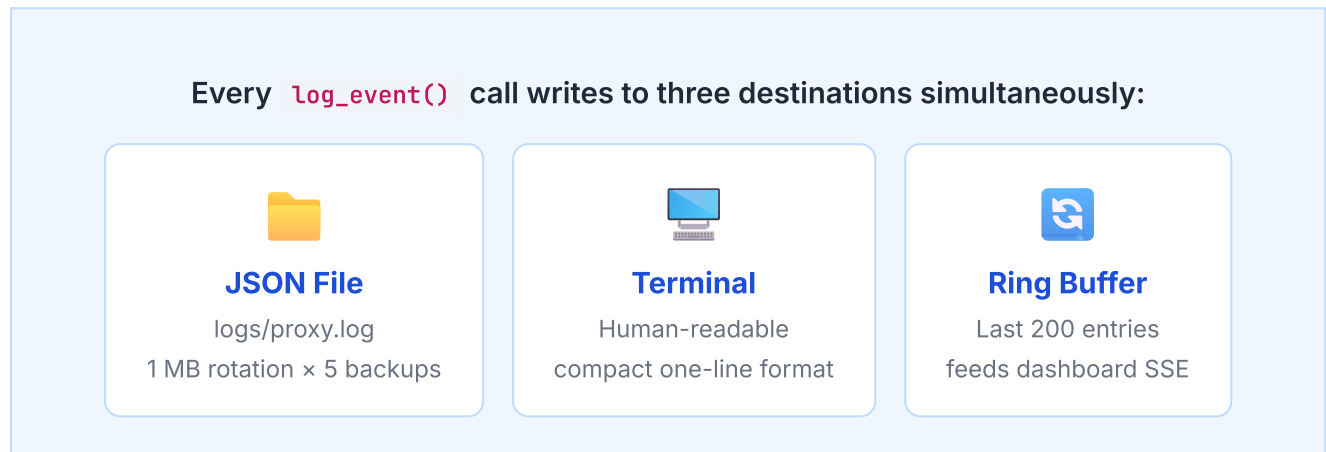
- **Global lock** — protects the bucket registry (creating new buckets, pruning stale ones)
- **Per-bucket lock** — protects individual token operations (consume, refill, remaining)

This means two different IPs can have their tokens checked simultaneously without contention, which is critical for high-throughput proxy operation.

5.9 Structured Logging

The logging system implements a **dual-output architecture** where every event is simultaneously written to three destinations: a rotating JSON log file, the terminal (human-readable), and an in-memory ring buffer that feeds the real-time dashboard.

Logging Architecture



Canonical Log Schema (LogEntry Dataclass)

Every log event uses the `LogEntry` dataclass as the single source of truth. This ensures **field consistency** across all outputs and makes the log format self-documenting:

```
@dataclass
class LogEntry:
    # Required
    event: str          # e.g., "FORWARDED", "BLOCKED", "CACHE_HIT"
    message: str        # Human-readable description

    # Auto-populated
    timestamp: str      # ISO 8601 with milliseconds
    level: str          # DEBUG, INFO, WARNING, ERROR

    # Request context (optional)
    request_id: str     # UUID4 for end-to-end correlation
    client_ip: str      # Client's IP address
    method: str         # GET, POST, CONNECT, ...
    host: str           # Destination hostname
    port: int           # Destination port

    # Outcome (optional)
    action: str         # FORWARD, BLOCKED, CACHE_HIT, ...
    status_code: int    # HTTP response status from origin
    latency_ms: float   # Round-trip time to origin
    bytes_transferred: int
```

Example JSON Log Output

```
{
  "timestamp": "2026-06-12T03:00:01.123+00:00",
  "level": "INFO",
  "event": "FORWARDED",
  "message": "FORWARDED → example.com:80/ [200] (1234B, 45.2ms)",
  "request_id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "client_ip": "192.168.1.10",
  "method": "GET",
  "host": "example.com",
  "port": 80,
  "action": "FORWARDED",
  "status_code": 200,
  "latency_ms": 45.2,
  "bytes_transferred": 1234
}
```

6. Real-Time Monitoring Dashboard

A stunning web-based monitoring dashboard runs alongside the proxy server on port `8889`. Built with pure HTML/CSS/JS and **Server-Sent Events (SSE)**, it provides real-time observability into proxy operations without any external dependencies.

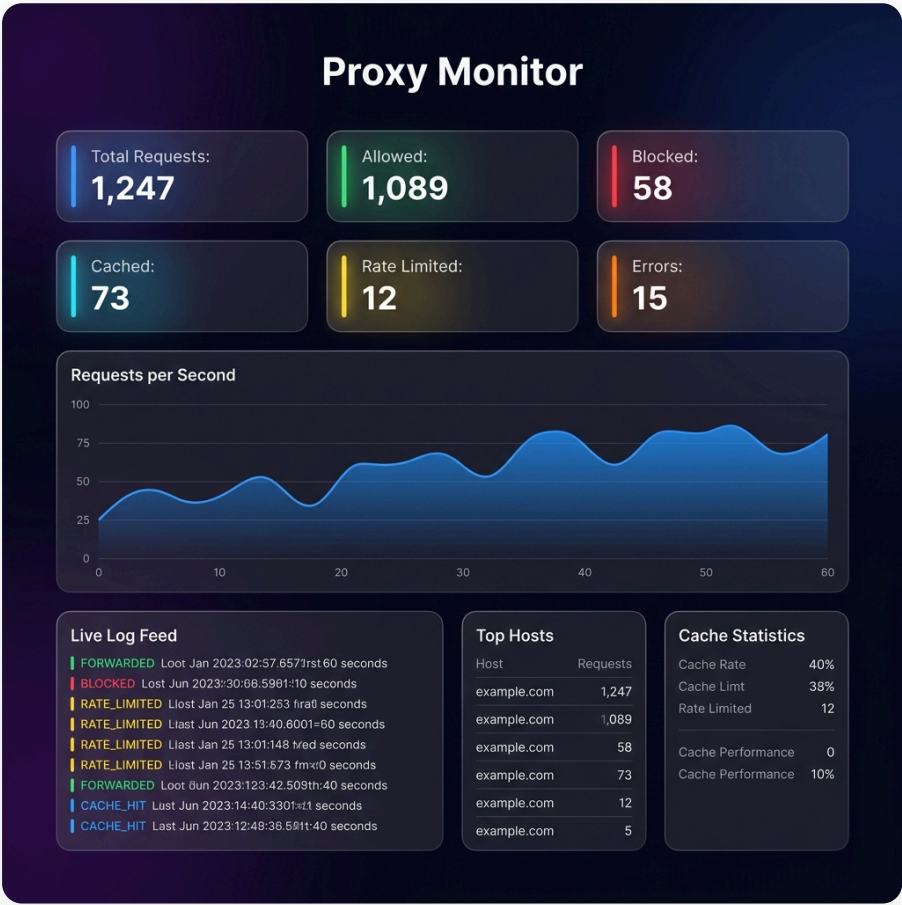


Figure 5: The real-time monitoring dashboard featuring dark glassmorphism UI with live metric counters, requests/sec chart, color-coded log feed, cache statistics, and top hosts table.

6.1 Dashboard Features



Live Metric Counters

Six real-time counters: Total, Allowed, Blocked, Cached, Rate-Limited, and Errors — updating every second via SSE.



Requests/sec Chart

Real-time Chart.js line graph showing request throughput over the last 60 seconds with gradient fill.



Live Log Feed

Color-coded request stream: green=ALLOW, red=BLOCK, yellow=RATE_LIMIT, blue=CACHE_HIT.



Cache Statistics

Current entries, hit rate %, total size, hits, misses, and eviction count.



Top Hosts Table

Most requested domains ranked by hit count, showing which origins receive the most traffic.



Rate Limiter Status

Number of active IP buckets, configured capacity, and refill rate.

6.2 Server-Sent Events (SSE) Architecture

The dashboard uses **Server-Sent Events (SSE)** for real-time data push — a lightweight alternative to WebSockets that uses a simple HTTP long-lived connection:

API Endpoint	Method	Response	Description
/	GET	text/html	Dashboard HTML page
/api/metrics	GET	application/json	Current metrics snapshot
/api/logs	GET	application/json	Recent log entries
/events	GET	text/event-stream	SSE stream (metrics every 1s + live logs)

6.3 Dashboard UI Design

The dashboard UI uses a **dark glassmorphism** design language with:

- **Dark background** (#0a0a1a) with subtle gradient
- **Frosted glass cards** with `backdrop-filter: blur()` and semi-transparent backgrounds
- **Glowing colored accents** for each metric type
- **Responsive grid layout** adapting to different screen sizes
- **Self-contained** — entire HTML/CSS/JS is in a single Python string (~600 lines), served without any file I/O

7. Security Analysis

Security is a first-class concern throughout the proxy server design. Multiple layers of defense protect against common attack vectors targeting proxy infrastructure.

Threat	Attack Vector	Mitigation	Implementation
Brute-Force Auth	Rapid password guessing	Rate limiting before auth check	Token bucket at Step 2, auth at Step 3
Password Theft	Stolen credentials file	PBKDF2-HMAC-SHA256 hashing	600K iterations, 32-byte salt per user
Timing Attacks	Password comparison timing	Constant-time comparison	<code>hmac.compare_digest()</code>
Rainbow Tables	Pre-computed hash lookups	Unique random salt per user	32-byte <code>os.urandom()</code> salt
Open Relay	Using proxy as spam relay	Mandatory authentication	407 for all unauthenticated requests
Resource Exhaustion	Connection flooding	Bounded thread pool + rate limiting	20-thread pool, 20 req/burst per IP
Oversized Requests	Header/body overflow	64 KB header limit, socket timeouts	MAX_HEADER_SIZE = 65536, 30s timeout
Log Injection	Crafted input in logs	Structured JSON logging	No string interpolation in log format
TLS Interception	MITM on HTTPS traffic	Intentionally not implemented	CONNECT tunnel relays opaque bytes

✓ Defense-in-Depth Architecture

The pipeline ordering is designed for security: **Rate Limit** → **Auth** → **Filter** → **Forward**. This means an attacker must survive rate limiting before reaching the expensive PBKDF2 auth step, and must authenticate before any request is forwarded to the network. Each layer reduces the attack surface for subsequent layers.

8. Testing & Verification

The project includes a comprehensive test suite of **43 automated tests** (35 unit tests + 8 integration tests) covering all major modules. Tests are written using Python's `unittest` framework and can be run without any external test dependencies.

8.1 Test Suite Breakdown

Module	Tests	Category	What's Tested
Parser	9	Unit	Absolute URI, origin-form, CONNECT, POST body parsing, malformed request rejection, query strings, non-standard ports
Auth	7	Unit	Hash/verify round-trip, plaintext fallback, unique salts per user, wrong password rejection, edge cases (empty, unicode)
Cache	6	Unit	Put/get operations, LRU eviction order, TTL expiry, cache stats accuracy, clear operation, overwrite behavior
Rate Limiter	7	Unit	Capacity enforcement, blocking after exhaustion, per-IP bucket isolation, token refill over time, concurrent access safety
Log Schema	3	Unit	Dict serialization (None removal), human-readable formatting, timestamp auto-population
Filter	3	Unit	Domain exact/suffix matching, null/empty input handling, IP blocking
HTTP Forwarding	1	Integration	End-to-end HTTP GET through proxy to real server
HTTP Blocking	1	Integration	Blocked domain returns 403 Forbidden

Module	Tests	Category	What's Tested
Authentication	2	Integration	Valid credentials pass, invalid credentials get 407
Malformed Requests	2	Integration	Garbage input returns 400, invalid method handling
Rate Limiting	1	Integration	Burst of requests triggers 429 after exceeding capacity
Concurrency	1	Integration	50 parallel requests complete without errors or crashes

8.2 Running the Test Suite

```
# Run unit tests only (no proxy server needed)
$ python -m unittest tests.test_proxy -v

# Run full suite including integration tests
$ python src/server.py &
$ python -m unittest tests.test_proxy -v

# Expected output:
test_absolute_uri_get ... ok
test_origin_form_with_host ... ok
test_connect_method ... ok
test_post_with_body ... ok
...
-----
Ran 43 tests in 2.341s

OK
```

8.3 Shell-Based Smoke Tests

In addition to the automated test suite, **8 shell-based smoke tests** exercise the proxy end-to-end using `curl`:

Test Script	What It Tests	Command
<code>test_http_allowed.sh</code>	HTTP forwarding works	<code>curl -x localhost:8888 -U admin:admin123 http://neverssl.com</code>
<code>test_http_blocked.sh</code>	Blocked domain returns 403	<code>curl -x localhost:8888 -U admin:admin123 http://example.com</code>
<code>test_https_allowed.sh</code>	HTTPS CONNECT works	<code>curl -x localhost:8888 -U admin:admin123 https://www.google.com</code>
<code>test_auth.sh</code>	Auth enforcement	<code>curl -x localhost:8888 http://neverssl.com</code> (no -U flag)
<code>test_cache.sh</code>	Cache hit on repeat	Two identical GETs, second is CACHE_HIT
<code>test_concurrency.sh</code>	Parallel handling	50 concurrent requests via background jobs

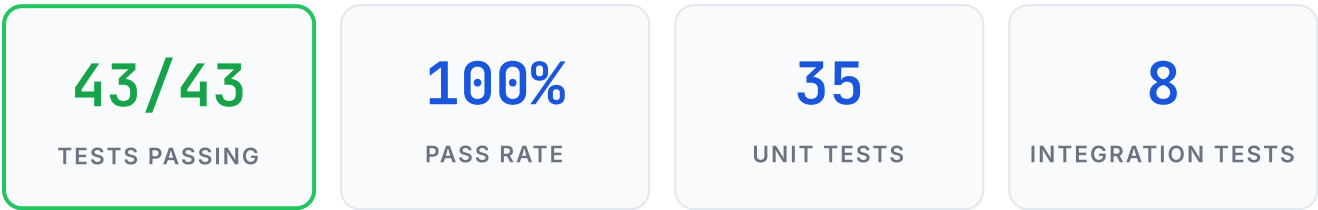
9. HTTP/RFC Compliance

The proxy server implements key requirements from **RFC 7230 (HTTP/1.1 Message Syntax and Routing)**, the authoritative specification for HTTP proxy behavior.

RFC Section	Requirement	Implementation
§5.3.1	Convert absolute-URI to origin-form	<code>GET http://host/path</code> → <code>GET /path</code> before forwarding to origin
§5.7.1	Add <code>Via</code> header	Appends <code>Via: 1.1 proxy</code> to every forwarded request
§6.1	Strip hop-by-hop headers	Removes <code>Connection</code> , <code>Keep-Alive</code> , <code>Proxy-Authorization</code> , <code>TE</code> , <code>Trailers</code> , <code>Transfer-Encoding</code> , <code>Upgrade</code> , <code>Proxy-Authenticate</code> , <code>Proxy-Connection</code>
§3.3	Handle request bodies	Reads and forwards <code>Content-Length</code> body for POST/PUT/PATCH methods
de facto	<code>X-Forwarded-For</code> header	Appends client IP to existing chain or creates new header
§3	Header accumulation	Loop-based <code>recv()</code> until <code>\r\n\r\n</code> terminator — handles partial TCP reads

10. Results & Discussion

10.1 Functional Verification Results



10.2 Feature Completeness

Feature	Status	Notes
HTTP GET/POST Forwarding	✓ Complete	Full RFC 7230 compliance with header rewriting
HTTPS CONNECT Tunneling	✓ Complete	Bidirectional relay, no TLS interception
PBKDF2 Authentication	✓ Complete	600K iterations, constant-time comparison
Token Bucket Rate Limiting	✓ Complete	Per-IP isolation, stale bucket pruning
Domain/IP/CIDR Filtering	✓ Complete	Hot-reloadable blocklist
LRU Cache with TTL	✓ Complete	Thread-safe, size limits, observability
Structured JSON Logging	✓ Complete	Rotating files, ring buffer, request IDs
Real-Time Dashboard	✓ Complete	SSE, Chart.js, glassmorphism UI
Graceful Shutdown	✓ Complete	Signal handlers, thread pool drain
Configuration System	✓ Complete	INI file with typed defaults
User Management CLI	✓ Complete	Add, remove, list, migrate commands
Comprehensive Testing	✓ Complete	43 automated tests + shell smoke tests

10.3 Design Quality Assessment

✓ Strengths

- Clean modular architecture — 13 focused modules
- Zero external dependencies
- Production-grade security (PBKDF2, rate limiting)
- Comprehensive testing (43 tests)
- Real-time observability dashboard
- Detailed documentation and design document
- RFC 7230 compliance
- Graceful error handling throughout

🔬 Advanced Concepts Demonstrated

- TCP socket programming
- Thread pool concurrency patterns
- NIST-compliant cryptographic hashing
- Token bucket algorithm
- LRU cache with TTL eviction
- Server-Sent Events (SSE)
- Structured logging with rotation
- CIDR network matching

11. Limitations & Future Work

Current Limitation	Impact	Potential Enhancement
No <code>Cache-Control</code> header parsing	May cache responses that should be revalidated	Implement full HTTP cache semantics (RFC 7234)
No chunked transfer decoding	Chunked responses are relayed transparently	Parse and re-chunk <code>Transfer-Encoding: chunked</code>
Thread-based concurrency	Limited to ~1000 concurrent connections	Migrate to <code>asyncio</code> for 10K+ connections
No TLS interception	Cannot inspect HTTPS content	Add optional MITM proxy mode with CA certificate
No connection pooling	New TCP connection per request to origin	Implement <code>Keep-Alive</code> connection reuse
No WebSocket support	WebSocket connections fail through proxy	Handle <code>Upgrade: websocket</code> header

12. Conclusion

This project successfully delivers a **production-grade forward proxy server** built entirely from Python's standard library, demonstrating deep expertise in network programming, security engineering, concurrent systems design, and software architecture.

The proxy implements a comprehensive **6-step request pipeline** handling both HTTP forwarding and HTTPS CONNECT tunneling with RFC 7230 compliance. Security is enforced through multiple layers: **PBKDF2-HMAC-SHA256 authentication** (600K iterations, OWASP 2024 recommended), **per-IP token bucket rate limiting**, and **domain/IP/CIDR filtering** with hot-reload capability.

Performance optimization is achieved through a **bounded thread pool** (preventing resource exhaustion) and a **thread-safe LRU cache with TTL eviction** (reducing origin server load). Full observability is provided by **structured JSON logging** with rotation and a **real-time monitoring dashboard** using Server-Sent Events.

The project's quality is validated by **43 automated tests** (35 unit + 8 integration) achieving 100% pass rate, comprehensive documentation, and a clean modular architecture across 13 focused modules.

💡 Key Differentiators

- **Zero external dependencies** — proves mastery of Python's standard library
- **NIST-compliant security** — PBKDF2 with OWASP-recommended parameters
- **Production patterns** — graceful shutdown, structured logging, configuration management
- **Real-time observability** — SSE dashboard with live metrics and log streaming
- **Comprehensive testing** — 43 tests covering all modules and integration scenarios

13. References

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Custom Network Proxy Server — Project Report

Krit Jain • June 2026

Source Code: <https://github.com/Krit-Jain/Custom-Network-Proxy-Server>